INCH-POUND
MIL-DTL-1947B(SH)
17 March 2023
SUPERSEDING
MIL-B-1947A(SHIPS)
18 August 1960

DETAIL SPECIFICATION

SOOT BLOWERS (SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steam boiler soot blowers intended for cleaning boiler firesides.

1.2 Classification. Soot blowers are of the following types and classes, as specified (see 6.2).

1.2.1 Types. Refer to 6.8 for a cross-reference of old to new type designations and substitutability data.

a. Type I – Stationary.

b. Type II – Rotary.

c. Type III – Long retractable.

1.2.2 Classes.

a. Class 1 – Maximum boiler operating pressure 710 pounds per square inch (psi); maximum steam temperature 875 °F.

b. Class 2 – Maximum boiler operating pressure 1500 psi; maximum steam temperature 975 °F.

1.3 Part or identifying number (PIN). The PINs to be used for soot blowers acquired to this specification are created as follows:

M	**1947**	**-**	**X**	**X**
Prefix for military specification	Specification number	-	Type (see code below)	Class (see code below)

PIN codes				
Type	**Code**		**Class**	**Code**
I	A		1	1
II	B		2	2
III	C			

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AMSC N/A FSC 4410

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Example PIN: M1947-C2 is a long retractable soot blower (type III) for maximum operating pressure and temperature of 1500 psi and 975 °F (class 2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

FF-N-836 - Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat

FF-S-85 - Screw, Cap, Slotted and Hexagon Head

FF-S-86 - Screw, Cap, Socket-Head

FF-S-92 - Screw, Machine, Slotted, Cross-Recessed or Hexagon Head

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-22 - Welded Joint Design

MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships

MIL-STD-2035 - Nondestructive Testing Acceptance Criteria

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume 1

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

321-8860241 - Soot Blower System Remote Closing Power System Installation

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804-841336 - Piping, Boiler Soot Blower

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the TMIN without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://webstore.ansi.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS ½ through NPS 24, Metric/Inch Standard

Boiler and Pressure Vessel Code, Section VIII - Rules for Construction of Pressure Vessels Division 1

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A106/A106M - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service

ASTM A182/A182M - Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service

ASTM A217/A217M - Standard Specification for Steel Castings, Martensitic Stainless and Alloy, for Pressure-Containing Parts, Suitable for High-Temperature Service

ASTM A276/A276M - Standard Specification for Stainless Steel Bars and Shapes

(Copies of these documents are available online at www.astm.org.)

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SAE INTERNATIONAL

SAE AMS5387 - Alloy, Corrosion and Heat Resistant, Investment Castings 60Co – 29Cr – 4.5W as Cast

SAE J1086 - Numbering Metals and Alloys

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Head design characteristics. For the head materials specified for each class of soot blowers (see 3.6.3.6.1 and 3.6.3.6.2), actual maximum steam temperature may exceed the specified maximum material temperature by not more than 25 °F.

3.3 Mounting flanges. Class 1 soot blowers shall be furnished with ASME raised-face standard mounting flanges in accordance with ASME B16.5 and the Boiler and Pressure Vessel Code, Section VIII, for the maximum pressure and temperature specified. Class 2 soot blowers shall be furnished with 1500 psi ASME raised-face standard flanges in accordance with ASME B16.5. All head inlets shall be flanged. If the flanged inlet provides an access opening to valve parts, an adapter mounting flange to ASME standards shall be furnished. The gasket contact surfaces of all flanges used in the soot blower assembly shall be finished with a circular lay (concentric or phonographic) having a roughness not exceeding 500 roughness average (Ra) produced by machining cuts of 0.003-inch maximum depth but not less than 40 cuts per inch of face width.

3.4 Materials. Materials used in the construction of soot blowers for naval ships, where not specified herein, shall be the lightest practicable, consistent with the strength required for safety and reliability, temperature, and pressure requirements herein.

3.4.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4.2 Threaded parts.

3.4.2.1 Threaded standards. Standards for threaded parts shall be in accordance with FED-STD-H28.

3.4.2.2 Standard bolts, nuts, and machine screws. Unless otherwise specified (see 6.2), all bolts and nuts shall be in accordance with MIL-DTL-1222. Unless otherwise specified (see 6.2), machine screws shall be in accordance with FF-S-85, FF-S-86, FF-S-92, and FF-N-836, as applicable. Nuts and bolts on moving parts or subject to vibration during operation shall be secured by a locking device.

3.4.2.3 Thread projection. Approximately one thread shall project clear from the outer face of the nut when nuts are screwed home. Minimum thread protrusion is where the leading edge of the first thread on the bolt or stud end is flush with the face of the nut.

3.5 Component parts. Soot blowers shall consist of the following parts:

a. Operating mechanism (see 3.6).
b. Element or lance (see 3.7).
c. For type I and II soot blowers only, bearings or hanger (see 3.8).
d. Boiler casing fitting (stuffing box or boiler casing seal [see 3.9]).

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3.6 Operating mechanism.

3.6.1 Description. Soot blower operating mechanism for each type shall consist of the following:

a. Type I (stationary) operating mechanism shall consist of a steam admission device (valve) only.

b. Type II (rotary) operating mechanism shall consist of a steam head or housing, including mechanical means of element rotation geared into the steam admission device.

c. Type III (long retractable) operating mechanism shall consist of the following:

(1) Steam head to regulate steam admission device.

(2) Traveling carriage to effect lance translation and rotation.

3.6.2 Type I (stationary) soot blowers (operating mechanism). Type I (stationary) soot blowers shall include an adjustable orifice that is preset to 300 psi (or less as approved by the Technical Authority [see 6.7]) steam admission to the element. This orifice installation, or other means to be installed in the vicinity, shall contain a ⅜-inch national pipe thread (NPT) tap for the insertion of a pressure-measuring device to verify admission steam pressure entering the element. Unless otherwise specified (see 6.2), type I soot blowers shall be complete with steam valves. Steam valves shall be in accordance with MIL-STD-777 material for the source of steam. A spring-loaded air relief valve of corrosion-resistant material (⅜-inch nominal pipe size [nps] minimum) shall be provided to admit purging air to the element and to prevent combustion gases from entering the piping. A ⅜-inch connection shall be provided for this purpose. The air relief valve shall be provided with fittings for connection to the pressurized air space by ½-inch steel or copper tubing. Unless otherwise specified (see 6.2), type I soot blowers shall be complete with air relief tubing.

3.6.3 Steam head (types II and III).

3.6.3.1 Operation. The steam head shall be designed to transmit steam to the element or lance from a fixed steam connection without steam leakage when the element or lance is being operated. It shall be designed so that the head inlet is on or below the horizontal in a plane normal to the axis of the soot blower element. A typical arrangement for soot blower piping is shown in 804-841336.

3.6.3.2 Steam admission valve. The steam admission valve shall be either power operated by an air motor or manually operated, as specified (see 6.2). Manual operation may be either by means of a single endless chain or by a crank as approved by the Technical Authority. Extension rods and gearing may be substituted if space limitations so indicate. An operation instruction plate shall be provided (see 3.10).

3.6.3.3 Steam pressure control in head. The method of steam admission and steam pressure control shall be as approved by the Technical Authority. Means of pressure control shall be included in each head design to reduce the pressure at the element to a maximum blowing pressure of 300 psi. A ⅜-inch NPT tap is to be provided in each soot blower head for insertion of a pressure-measuring device. This pressure control shall be an adjustable part of the head in order to affect no greater steam entrance pressure to the element than 300 psi. The control shall be capable of external adjustment and shall be preset to the designated maximum.

3.6.3.4 Power drive. When air-motor-driven soot blowers are specified, these air motors shall operate both the steam admission valves and the designated movement of the elements. The use of cast iron in the motors is not permitted. The drive shall also include the necessary pushbutton stations for the actuation of the air motors. Panel boards for mounting the pushbutton stations, including remote shutdown at the enclosed operating station, shall be provided for locations and functions in accordance with 321-8860241. The panel boards shall have an interlocking device to prevent operation of more than one soot blower at a time and an indicating light to show when a soot blower is in operation. Interconnecting piping between air motors and control panels shall not be furnished. Plates with operation instructions, including hand operation, shall be provided at each panel and operating location in accordance with 3.10.

3.6.3.5 Scavenging air valve. A spring-loaded scavenging air valve (see 3.6.2) shall be provided on the soot blower head to provide purging air to the element and to prevent combustion gases from entering the soot blower head. A ⅜-inch connection on the soot blower head shall be provided for this purpose. Unless otherwise specified (see 6.2), the scavenging air relief valves shall be connected to the pressurized air space by ½-inch steel or copper tubing and by joints approved by the Technical Authority.

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3.6.3.6 Head materials. Minimum design requirements for head materials shall be as specified in 3.6.3.6.1 and 3.6.3.6.2. All head castings shall include a head marking in a location observable by the operating personnel. Markings on class 1 head castings shall read "710" in figures at least ½ inch in height. Markings on class 2 head castings shall read "1500" in figures at least ½ inch in height. Class 2 head materials are acceptable for class 1 use as approved by the Technical Authority.

3.6.3.6.1 Class 1, 710 psi operating pressure, 875 °F. Minimum class 1 design requirements for head materials shall be as follows:

a. Head – Carbon molybdenum alloy steel castings or forgings in accordance with ASTM A217/A217M, grade WC1 (UNS J12524[1]) or ASTM A182/A182M, grade F1 (UNS K12822), respectively.

b. Valve discs – Cast alloy steel in accordance with any grade of ASTM A217/A217M or forged alloy steel in accordance with any grade of ASTM A182/A182M or any grade of ASTM A276/A276M. The seating area shall be faced with a ring of cobalt alloy in accordance with SAE AMS5387 (commercially known as Stellite 6, UNS R30006), in a layer not less than 3/32 inch thick.

c. Valve seats – Materials shall be the same as for the discs. Seats may be made an integral part of the body or may consist of tubular insert welded or heat shrunk into the body. The seating area shall be faced with cobalt chromium alloy in accordance with SAE AMS5387 (UNS R30006), in a finished layer not less than 3/32 inch thick.

d. Valve stems – Martensitic corrosion-resistant steel in accordance with ASTM A182/A182M or ASTM A276/A276M.

e. Snap rings – When used, snap rings shall be made of 17-7 precipitation-hardening (PH) stainless steel, condition C (UNS S17700).

3.6.3.6.2 Class 2, 1500 psi operating pressure, 975 °F. Minimum class 2 design requirements for head materials shall be as follows:

a. Head – 1¼ percent chrome; ½ percent molybdenum alloy steel casting or forging in accordance with ASTM A217/A217M, grade WC6 (UNS J12072) or ASTM A182/A182M, grade F11 (UNS K11572), respectively.

b. Valve discs – Cast chromium alloy steel in accordance with ASTM A217/A217M, grade WC6 (UNS J12072) or forged chromium alloy steel in accordance with ASTM A182/A182M, grade F11 (UNS K11572). The seating area shall be faced with a ring of cobalt alloy in accordance with SAE AMS5387 (UNS R30006), in a layer not less than 3/32 inch thick.

c. Valve seats – Valve seats shall be of the same material as the valve discs. Valve seats shall be made as above for class 1 heads. The seating area shall be faced with cobalt alloy in accordance with SAE AMS5387 (UNS R30006), in a finished layer not less than 3/32 inch thick.

d. Valve stems – Martensitic corrosion-resistant steel in accordance with ASTM A182/A182M or ASTM A276/A276M.

e. Snap rings – When used, snap rings shall be made of 17-7 PH stainless steel, condition C (UNS S17700).

3.6.4 Steam head for type II soot blowers.

3.6.4.1 Operation. In addition to the operation requirements of 3.6.3, type II soot blower steam heads shall be provided with means for rotating the element in conjunction with the operation of the steam admission valve. Direction of operation in a clockwise or counterclockwise direction shall be as specified (see 6.2) based on boiler requirements to keep header and casing-pocket areas clean and direct loosened soot into uptake gas paths. The provided total or combined angle or rotation of the element to fully open and to fully close the steam admission valve shall not exceed 60 degrees of arc.

[1] Unified Numbering System (UNS) alloy numbers per SAE J1086.

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3.6.4.2 Reduction gearing. Reduction gearing shall be provided in the head for rotation of the elements. The gear ratio shall be not less than seven to one. For manually operated units designed for rotation in one direction only, an arrow shall be cast on or attached to the gear wheel, sheave wheel, or operating handwheel, whichever is most observable by the operating personnel, to indicate the correct direction of rotation for the operation of the blower. For power-operated units, the correct direction of rotation for the element shall be indicated on the unit wherever it is observable.

3.6.4.3 Connecting piece (swivel or extension tube). A tubular connecting piece shall be provided between the head and the element. Corrosion-resistant material shall be used where the piece passes through the stuffing box in the head. The connecting piece shall be for the purpose of securing the element to the head during field installation. The connecting piece shall ensure maintenance of the proper angular position of the nozzles with respect to the operating mechanism in the head. An indicator to show the angular position of the nozzles to the operator at all times shall be attached to the gear wheel or secured by other means approved by the Technical Authority. The swivel tube or the element shall be at least 6 inches longer than the designed length to allow for field modification to suit installation conditions.

3.6.4.4 Head interchangeability. All heads of like design within the same class shall have fully interchangeable internal parts.

3.6.5 Operating mechanism for type III soot blowers. Operating mechanism shall consist of:

a. A steam head containing a poppet valve with an externally adjustable pressure control of the steam admission source.

b. A traveling carriage mechanism that shall be air motor driven. Mechanical means shall be used to translate and rotate the lance. Manual means shall be provided to allow the removal of the lance from boiler cavities in case of air motor failure. For manual emergency retraction of the lance, in addition to gearing with the handwheel, means shall be furnished to allow manual rotation of the gearing into an accessible position for emergency retraction. The traveling carriage shall be of rugged construction, with means of supporting the lance to prevent whipping during the rotation and translation of operation. Minimum rotation speed of the lance shall be five revolutions per minute. Minimum translation speed shall be 5 feet per minute. Means shall be provided to mechanically restrain the feed tube from rotation in the steam chest. In addition, the feed tube shall be ground and polished to reduce wear on the packing. Means shall be provided to prevent, under all conditions of boiler or soot blower operation, any gas leakage around the lance where it passes through the boiler casing. Instruction plates shall be provided for carriage operation (see 3.10).

3.7 Element or lance. Soot blower elements (or lances) shall be steam tubes projecting into the boiler firesides. Each element shall have nozzle outlets that discharge steam at a high velocity for loosening soot so it can be blown out of the boiler.

3.7.1 Operation. Elements shall be rotary, stationary, or retractable. Rotary elements or lances shall be operated by a steam head (see 3.6). Stationary elements shall be operated by an externally adjustable steam throttling valve. Long retractable elements shall have a traveling carriage and an enclosed poppet valve.

3.7.2 Grades. Elements shall be of the following grades:

Grade	Maximum gas temperature, °F
I	900
II	1500
III	1800
IV	over 1800

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3.7.3 Material (elements). Grades I and II elements shall be manufactured of seamless rolled or drawn carbon steel pipe in accordance with ASTM A106/A106M (UNS K03006). The material of the grade II elements shall be completely protected against oxidation by an aluminum coating properly impregnated, or by a chromium calorizing or plating process, the coating to be applied after installation of the nozzles. The use of grade I and grade II element materials at a lower temperature than that approved by the Technical Authority shall be allowed, provided the physical properties of the material at the lower temperature meet the purpose. Grades III and IV elements shall be fabricated of corrosion-resistant steel alloys; grade of alloy and need of protective coating shall be as approved by the Technical Authority. All welding of elements shall be in accordance with MIL-STD-22 and S9074-AR-GIB-010/278.

3.7.4 Design. The wall thickness of the element, when manufactured of low carbon steel, shall be at least equal to that of a schedule 80 pipe. When chrome alloy steel elements are used, the thickness of the wall of the element shall be as approved by the Technical Authority. The end of the element opposite the soot blower head shall be closed in a manner approved by the Technical Authority. This closure shall not be accomplished by use of a cap or plug attached by screw threads.

3.7.5 Elements for types I and II soot blowers.

3.7.5.1 Multi-nozzle element. A multi-nozzle element shall be a straight pipe with at least one row of small nozzles fitted along one side. The element shall be either (a) rotary (with a steam valve and gearing parts of the soot blower head) or (b) stationary (with a steam valve only).

3.7.5.2 Adjustment of the element. Elements shall be fabricated to exact designed lengths as specified (see 6.2) for the intended boiler locations. Accurate longitudinal adjustment of the element in the field to ensure correct position of the nozzles relative to the boiler tubes, when blowing, is accomplished by field modification (cutting and threading element or swivel tube or adjusting support bar and carriage). Couplings or flanges used between the element and the swivel or extension tube shall be threaded to a minimum such that when the element is also threaded to a minimum and in the correct position, at least ¾ inch of the element will be engaged.

3.7.5.3 Multi-nozzle design. Nozzles shall be designed to provide maximum steam velocity. Material of nozzles shall be the same or of superior mechanical properties than that of the element material. The diameter of the nozzle throat shall be sized as required by the Technical Authority (see 6.2). Unless otherwise approved by the Technical Authority, the ratio of the combined throat areas of all nozzles installed in a soot blower element to the internal cross-sectional area of the element shall not exceed 1.2 to 1. Unless otherwise approved by the Technical Authority, the maximum overall length of the nozzle shall be ⅝ inch. The nozzles shall be secured by welding or by screw threads and welding. Unless otherwise approved by the Technical Authority, all nozzles shall be installed such that the outlet end of the nozzle is flush with the outside of the element. There shall be only one row of nozzles in each element unless otherwise approved by the Technical Authority.

3.7.5.4 Position of nozzles. The spacing of the nozzles shall be as approved by the Technical Authority for each boiler class application, but in no case shall the spacing of the nozzles be greater than every third tube space of the row of tubes adjacent to the element when the axis of the element is at a right angle to the axis of the tubes; or greater than 3 inches when the axis of the element is parallel to the axis of the tubes. In cases where tubes of 2-inch diameter and larger are employed and the axis of the soot blower element is at a right angle to the axis of the tubes, the spacing of the nozzles shall be not greater than every tube space to secure a nozzle spacing of 3 to 3⅛ inches or as near thereto as practicable. Nozzles shall be arranged to blow into the lanes between the tubes and not against the tubes, insofar as is practicable. Except as specified herein, nozzles shall be installed with their axis normal to the axis of the element. When the axis of the element is at a right angle to that of the tubes, and the element is installed between two rows of tubes, staggered with respect to each other, if the tubes opposite the nozzles are 1½ inches or less from the outside of the element, the nozzles shall be angled not more than 10 degrees to prevent direct impingement of the steam on the tubes. When the axis of the element is parallel to that of the tubes and the nearest tubes are 1½ inches or less from the outside of the element, the nozzles shall be angled as described above.

3.7.6 Double nozzle lance for type III soot blower. The lance shall be ground and polished to prevent excessive binding or wear on the lance or related parts. Two opposed nozzles at an angle designed to suit the requirements shall be installed near the front end of the lance in accordance with 3.7.5.3.

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3.7.7 Gas temperatures above 1800 °F. When boiler furnace locations reaching maximum gas temperatures above 1800 °F require soot cleaning of the boiler firesides; retractable lances, multi-nozzle elements air cooled by gas baffles or air jackets, and higher temperature materials shall be as required by the Technical Authority (see 6.2).

3.8 Bearings or hangers (for type I and type II soot blowers only).

3.8.1 Purpose and position of bearings. Bearings shall be provided to support the soot blower element and to preserve alignment. Ample clearance shall be provided in the bearings for the elements to prevent seizing. The distance between bearings shall be not more than 2 feet 8 inches. When the axis of the element is at a right angle to that of the tubes, the bearing next to the boiler casing fitting shall be located with its center not more than six tubes from the end of the tube bank, and the projection of the element at the closed end beyond the last bearing shall be not more than 6 inches. When the axis of the element is parallel to that of the tubes, the element shall be supported as approved by the Technical Authority. Material of the bearings shall be the same as that of the element with which they are installed; slight variations in chemical characteristics due to a difference in fabrication methods is permitted.

3.8.2 Relation of bearings to tubes. The bearings shall be clamped to the tubes by U-bolts and nuts. When the slope of the tubes at the point of the attachment of the bearings is greater than 30 degrees above the horizontal, the bearings shall be double clamped to decrease the possibility of slipping. Bearings shall contact at least three tubes to prevent turning. The surfaces of bearings in contact with tubes shall be machined accurately to fit the circumference and curvature of the tubes to obtain the maximum contact surface and reduce the temperature of the bearings by conduction heat to the tubes.

3.9 Boiler casing fitting.

3.9.1 Boiler casing sleeve. A sleeve shall be provided to allow free passage of the element or lance through the boiler casing.

3.9.2 Boiler casing seal or stuffing box. The stuffing box shall be designed to permit passage of the soot blower element through the boiler outer casing and to effectively seal the sleeve opening against the leakage of combustion gases. This box shall also be designed to compensate for the relative movement between the connecting steam piping, the boiler tubes, and the boiler casing by a floating action that permits slight movement of the head in any direction in the plane normal to the axis of the element. Material for the stuffing box shall be either cast steel or ductile iron. Cast iron or malleable iron shall not be permitted. Stuffing boxes shall be flanged. Provisions shall be made for an air connection to the ½-inch outside diameter tubing with entry to the stuffing box through at least ⅜-inch NPT connections.

3.10 Marking. Identification and information plates and other markings shall be furnished in accordance with MIL-DTL-15024 and as a part of the machinery or equipment for which they are intended. Information plates containing instructions for operation and upkeep of the soot blowers shall be attached to boiler casings near the applicable soot blower type installations. Plate material shall be capable of withstanding 150 °F ambient temperature without damage.

3.11 Shock. Soot blowers shall withstand high-impact shock in accordance with MIL-DTL-901. Soot blowers shall be operable without steam or combustion gas leakage after shock exposure.

3.12 Reliability and maintainability. Each soot blower shall have an operating life of 100,000 hours with no scheduled maintenance. Corrective maintenance shall be no more than 2 hours per 20,000 hours of operation without the use of special tools. Special tools are defined as tools not listed in the GSA Global Supply Catalog.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 First article inspection. First article inspection shall be performed on one soot blower of each new design and construction when a first article sample is required (see 3.1). This inspection shall include the examinations and tests specified in table I.

4.3 Conformance inspection. Conformance inspection shall include the examinations and tests specified in table I on a sample selected in accordance with 4.3.2.

TABLE I. Examinations and tests.

Characteristics	Requirements	First article	Conformance
Design, materials, parts, mechanism, operation	All aspects of 3.2 through 3.10	4.4, 4.5.1, 4.5.2, 4.5.4.1, 4.5.4.2	4.4, 4.5.2
Scavenging air valve	3.6.3.5	4.5.3	4.5.3
Shock	3.11	4.5.5	-
Reliability and maintainability	3.6.4.4, 3.12	4.5.6	-

4.3.1 Lot. All soot blowers of the same class and type presented for delivery at one time shall be considered a lot for purposes of acceptance inspection. All parts required to be eventually assembled to form complete soot blowers shall be included in the lot.

4.3.2 Sampling for conformance inspection. A random sample of soot blowers shall be selected from each lot for the examinations specified in 4.4 in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.4 Examinations. Soot blowers shall be visually and dimensionally examined to determine conformance with the requirements of this specification as to materials, finish, workmanship, dimensions, components, fit, lubrication, marking, and identification and information plates. Any soot blower in the sample containing one or more material, visual, or dimensional defects shall be rejected, and if the number of defective soot blowers in any sample exceeds the acceptance number for that sample (see 4.3.2), the lot shall be rejected.

4.5 Tests and inspections. Soot blowers shall be tested and examined as follows. Failure to pass any test or examination shall be counted as a defect, and the soot blower shall be rejected. If the number of defective soot blowers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.5.1 Operation test. The soot blower shall be tested to determine effectiveness of steam control, economy in the use of steam, facility of operation, simplicity in installation, effectiveness in removal of soot, ruggedness of construction, quality of workmanship, and the suitability of the unit for the use specified. This test shall be demonstrated by an actual installation on a boiler of the same or similar design to those on naval ships.

4.5.2 Hydrostatic test. All parts of the soot blowers that are subject to line steam pressure shall be tested by hydrostatic pressure to 1½ times the maximum class pressure and at the class temperature (see 1.2.2).

4.5.3 Scavenging (air relief) valve hydrostatic test. The valve shall be tested by hydrostatic pressure to 200 psi.

4.5.4 Radiographic and magnetic particle inspections.

4.5.4.1 Radiographic inspection. All pressure-containing castings rated at 600 pounds and above shall be 100 percent radiographically inspected in accordance with T9074-AS-GIB-010/271 and MIL-STD-2035. Sections less than 1 inch in thickness shall meet class 1 requirements of MIL-STD-2035. Sections with thickness 1 inch and over shall meet class 2 requirements of MIL-STD-2035.

4.5.4.2 Magnetic particle inspection. All pressure-containing alloy forgings rated at 600 pounds and above shall be 100 percent magnetic particle inspected in accordance with T9074-AS-GIB-010/271. (Carbon steel pressure-containing forgings shall not be magnetic particle inspected.)

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4.5.5 Shock test. Components and assembled soot blowers shall be tested for high-impact shock in accordance with MIL-DTL-901.

4.5.6 Reliability and maintainability tests. The reliability of the soot blower shall be demonstrated by the successful completion of all examinations and tests herein and a reliability prediction. The maintenance procedures, removal of parts for repair, interchangeability of parts, and time required for accomplishment shall be verified during first article testing. The demonstration procedures in MIL-HDBK-470 may be used.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be useful, but is not mandatory.)

6.1 Intended use. The soot blowers specified herein are intended for cleaning boiler fireside soot accumulations on naval vessels propelled by conventional steam boilers. Current stock of soot blowers may be used until depleted; future requisitions for soot blowers should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Types and classes required (see 1.2).
c. When first article inspection is required (see 3.1).
d. Requirements for bolts, nuts, and screws if other than as specified herein (see 3.4.2.2).
e. If steam valves or air relief tubing is not required for type I soot blowers (see 3.6.2).
f. Whether air motor or manual means of operation are required for types II and III soot blowers (see 3.6.3.2).
g. Scavenging air valve joints and tubing if other than as specified herein (see 3.6.3.5).
h. Direction of rotation for type II soot blowers (see 3.6.4.1).
i. Lengths of types I and II soot blower elements (see 3.7.5.2).
j. Element nozzle throat diameter as required by the Technical Authority (see 3.7.5.3).
k. Element materials for temperatures above 1800 °F as required by the Technical Authority (see 3.7.7).
l. AQL, suggested is 1.0 (see 4.3.2).
m. Packaging requirements (see 5.1).
n. Requirements for documentation (see 6.4).
o. Requirements for repair parts (see 6.5 and 6.6).

6.3 Technical Authority approval and direction. Deviations from specified materials, procedures, and requirements, and selection of specific alternative materials and procedures require Technical Authority approval or direction. Requests should include supporting documentation.

6.4 Recommended documentation. The following documentation should be requested as specified (see 6.2):

a. Test reports.
b. Technical manuals.
c. Product drawings.

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d. Provisioning technical documentation.

e. Repair parts and tools.

6.5 Valve or head repair parts. One set of valve or head repair parts should be furnished as specified (see 6.2) for each ship on which the soot blowers are to be installed. Each set of repair parts should consist of the following, the quantity of which should be 10 percent of the total number of soot blowers installed on the ship:

a. Valve disc and stem sub-assembly.

b. Pinion gear.

c. Springs of each size used.

d. Swivel tube (if applicable to design).

e. Packing.

f. Flange gaskets.

g. Lance for type III soot blowers.

h. Element for hottest location in boiler.

6.6 Air motor repair parts. When air-drive operation is required, sufficient motor parts (bearings and similar repair parts) should be provided to maintain the motor for 1 year, as specified (see 6.2). Replacement heads of types II and III should be furnished with sufficient cam and trigger parts to allow their assembly to fit any rotation requirements, both 360 degrees and below. Replacement swivel tubes or elements supplied should provide sufficient length to allow field modification to actual boiler depth conditions (at least 6 inches). Replacement heads should allow attachment of either 1½-inch or 2-inch elements. Any other repair parts deemed necessary for 1 year's maintenance should be offered for specific approval by the Technical Authority.

6.7 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.8 Cross-reference of classifications and substitutability data. Type III short retractable soot blowers in accordance with MIL-B-1947A have been removed from this specification as they are no longer used on Navy ships. Type IV long retractable soot blowers have been redesignated as type III in revision B of this specification. As such, type III long retractable soot blowers in accordance with MIL-DTL-1947B are substitutable for type IV long retractable soot blowers in accordance with MIL-B-1947A.

6.9 Subject term (key word) listing.

Boiler

Cleaning

Steam

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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